Linear Algebra Review

Vector Spaces, Linear Maps, and the Spectral Theorem

Phase 0: Prerequisites Review | Document 2 of 3
Primary texts: Axler, *Linear Algebra Done Right*; Hoffman & Kunze, *Linear Algebra*

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

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Abstract

This document is a self-contained refresher of finite-dimensional linear algebra, covering the results needed for functional analysis, optimization theory, and machine learning. We develop vector spaces, linear maps, determinants, eigenvalue theory, inner product spaces, the spectral theorem for real symmetric matrices, quadratic forms, the singular value decomposition, matrix norms, and matrix calculus. Every theorem is stated precisely and proved in full. The reader who completes this review will have the algebraic foundations required for Phase 01 (functional analysis, optimization) and all subsequent phases.

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1 Introduction

This document is a refresher, not a first course. We assume the reader has seen linear algebra before but needs to recall the key definitions, theorems, and proof techniques. We are selective: only results that the rest of the curriculum depends on are included.

Here is what we cover and why each topic matters:

1. **Vector spaces** (Section 2). The axioms, subspaces, basis, and dimension form the vocabulary of all linear mathematics. Functional analysis (Phase 01) generalizes these to infinite-dimensional Banach and Hilbert spaces; the finite-dimensional case here provides the intuition and many of the proof templates.
2. **Linear maps** (Section 3). Null space, range, rank-nullity, and matrix representation. Neural networks are compositions of linear maps with pointwise nonlinearities; understanding the linear part is essential before studying the nonlinear approximation.
3. **Determinants** (Section 4). The determinant characterizes invertibility and appears in change-of-variables formulas for integration — critical for multivariate probability densities and normalizing flows.
4. **Eigenvalues and eigenvectors** (Section 5). Eigenspaces, algebraic and geometric multiplicity, the Cayley–Hamilton theorem, and full diagonalizability criteria. The spectral radius of a transition matrix governs mixing times of Markov chains; eigenvalue gaps control convergence rates of iterative algorithms.
5. **Inner product spaces** (Section 6). The Cauchy–Schwarz inequality, Gram–Schmidt orthogonalization, orthogonal complements, and the orthogonal decomposition theorem. These are the finite-dimensional versions of the Hilbert space results in Phase 01 and underlie least-squares estimation, PCA, and natural policy gradients.
6. **The spectral theorem** (Section 7). Real symmetric matrices are orthogonally diagonalizable. Positive definiteness, the Rayleigh quotient, the Courant–Fischer minimax theorem, and Cholesky factorization. Convexity of a twice-differentiable function is determined by positive semidefiniteness of its Hessian; the Fisher information matrix in natural policy gradient methods is symmetric positive semidefinite.
7. **Quadratic forms** (Section 8). The connection between symmetric matrices and quadratic functions is the bridge from linear algebra to optimization. Second-order Taylor expansion produces a quadratic form whose matrix is the Hessian.
8. **Singular value decomposition** (Section 9). Every matrix has an SVD. The Eckart–Young theorem gives optimal low-rank approximation. The pseudoinverse provides least-squares solutions. PCA is a direct application.
9. **Trace, matrix norms, and the matrix exponential** (Section 10). Trace identities, the operator and Frobenius norms, submultiplicativity, the condition number, and the matrix exponential. The trace is ubiquitous in matrix calculus; the matrix exponential solves linear ODEs and defines continuous-time Markov chains.
10. **Matrix calculus** (Section 11). Derivatives of scalar functions with respect to vectors and matrices. These are the computational tools needed to derive gradient updates in optimization and machine learning.

Why Linear Algebra Matters for AI/ML

Linear algebra is the computational backbone of machine learning. Functional analysis (Phase 01) generalizes vector spaces to infinite dimensions. Optimization theory uses eigenvalues and positive definiteness to characterize convex functions. Neural networks are compositions of linear maps with pointwise nonlinearities. The spectral theorem underlies PCA, covariance analysis, and the Fisher information matrix in natural policy gradient methods. Matrix calculus provides the derivatives needed for every gradient-

based algorithm.

2 Vector Spaces

Throughout, $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$.

Definition 2.1 (Vector Space)**.** A *vector space* over a field $\mathbb{F}$ is a set V together with operations of addition $+ : V \times V \to V$ and scalar multiplication $\cdot : \mathbb{F} \times V \to V$ satisfying:

1. **Commutativity:** $u + v = v + u$ for all $u, v \in V$.
2. **Associativity of addition:** $(u + v) + w = u + (v + w)$ for all $u, v, w \in V$.
3. **Additive identity:** There exists $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
4. **Additive inverse:** For each $v \in V$, there exists $w \in V$ with $v + w = 0$.
5. **Multiplicative identity:** $1 \cdot v = v$ for all $v \in V$.
6. **Associativity of scalar multiplication:** $a(bv) = (ab)v$ for all $a, b \in \mathbb{F}$, $v \in V$.
7. **Distributive laws:** $a(u + v) = au + av$ and $(a + b)v = av + bv$.

Example 2.1. Standard examples:

1. $\mathbb{F}^n = \{(x_1, \ldots, x_n) : x_i \in \mathbb{F}\}$ with componentwise operations.
2. $\mathbb{F}[x]_{\leq n}$, the polynomials of degree at most n, with the usual operations.
3. $C[a, b]$, the continuous functions $f : [a, b] \to \mathbb{R}$, with pointwise operations.
4. $\mathbb{R}^{m\times n}$, the $m \times n$ matrices with real entries.

2.1 Subspaces

Definition 2.2 (Subspace)**.** A subset $U \subseteq V$ is a *subspace* if U is itself a vector space under the same operations. Equivalently, U is a subspace if and only if:

1. $0 \in U$,
2. $u + v \in U$ for all $u, v \in U$ (closed under addition),
3. $\lambda u \in U$ for all $\lambda \in \mathbb{F}$, $u \in U$ (closed under scalar multiplication).

Proposition 2.1 (Intersection of Subspaces)**.** *If U_1, U_2 are subspaces of V, then $U_1 \cap U_2$ is a subspace of V.*

Proof. We verify the three conditions. (1) $0 \in U_1$ and $0 \in U_2$, so $0 \in U_1 \cap U_2$. (2) If $u, v \in U_1 \cap U_2$, then $u, v \in U_1$ so $u + v \in U_1$, and similarly $u + v \in U_2$, so $u + v \in U_1 \cap U_2$. (3) If $u \in U_1 \cap U_2$ and $\lambda \in \mathbb{F}$, then $\lambda u \in U_1$ and $\lambda u \in U_2$, so $\lambda u \in U_1 \cap U_2$. □

2.2 Sums and Direct Sums

Definition 2.3 (Sum of Subspaces)**.** If $U_1, \ldots, U_m$ are subspaces of V, their *sum* is

$$U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_i \in U_i\}.$$

Proposition 2.2. *The sum $U_1 + \cdots + U_m$ is the smallest subspace of V containing all the U_i.*

Proof. The sum is a subspace: it contains 0 (take each $u_i = 0$), is closed under addition (if $u = u_1 + \cdots + u_m$ and $v = v_1 + \cdots + v_m$ with $u_i, v_i \in U_i$, then $u + v = (u_1 + v_1) + \cdots + (u_m + v_m)$ with $u_i + v_i \in U_i$), and similarly for scalar multiplication. Any subspace containing all U_i must contain every sum $u_1 + \cdots + u_m$, so the sum is the smallest such subspace. □

Definition 2.4 (Direct Sum)**.** The sum $U_1 + \cdots + U_m$ is a *direct sum*, written $U_1 \oplus \cdots \oplus U_m$, if each element of the sum can be written in exactly one way as $u_1 + \cdots + u_m$ with $u_i \in U_i$.

Proposition 2.3 (Characterization of Direct Sums). *$V = U_1 \oplus U_2$ if and only if $V = U_1 + U_2$ and $U_1 \cap U_2 = \{0\}$.*

Proof. ($\Rightarrow$) If $v \in U_1 \cap U_2$, then $0 = v + (-v)$ where $v \in U_1$ and $-v \in U_2$. But also $0 = 0 + 0$ with $0 \in U_1$ and $0 \in U_2$. By uniqueness of the direct sum decomposition, $v = 0$.

($\Leftarrow$) Suppose $u_1 + u_2 = u_1' + u_2'$ with $u_i, u_i' \in U_i$. Then $u_1 - u_1' = u_2' - u_2 \in U_1 \cap U_2 = \{0\}$, so $u_1 = u_1'$ and $u_2 = u_2'$. □

Theorem 2.4 (Dimension of a Sum). *If U_1 and U_2 are finite-dimensional subspaces of V, then*

$$\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2).$$

Proof. Let $\{w_1, \ldots, w_k\}$ be a basis of $U_1 \cap U_2$. Extend it to a basis $\{w_1, \ldots, w_k, u_1, \ldots, u_p\}$ of U_1 and to a basis $\{w_1, \ldots, w_k, v_1, \ldots, v_q\}$ of U_2.

We claim $\{w_1, \ldots, w_k, u_1, \ldots, u_p, v_1, \ldots, v_q\}$ is a basis of $U_1 + U_2$. It spans $U_1 + U_2$ since any element of U_1 is a combination of the w's and u's, and any element of U_2 of the w's and v's.

For independence, suppose $\sum \alpha_i w_i + \sum \beta_j u_j + \sum \gamma_\ell v_\ell = 0$. Then $\sum \gamma_\ell v_\ell = -\sum \alpha_i w_i - \sum \beta_j u_j \in U_1$. But $\sum \gamma_\ell v_\ell \in U_2$, so $\sum \gamma_\ell v_\ell \in U_1 \cap U_2$, hence $\sum \gamma_\ell v_\ell = \sum \delta_i w_i$ for some scalars. Since the w's and v's are a basis of U_2 and hence linearly independent, all $\gamma_\ell = 0$ and all $\delta_i = 0$. Then $\sum \alpha_i w_i + \sum \beta_j u_j = 0$, and since the w's and u's form a basis of U_1, all $\alpha_i = 0$ and $\beta_j = 0$.

Therefore $\dim(U_1 + U_2) = k + p + q = (k + p) + (k + q) - k = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2)$. □

2.3 Span and Linear Independence

Definition 2.5 (Span). The *span* of a list of vectors $v_1, \ldots, v_m \in V$ is

$$\operatorname{span}(v_1, \ldots, v_m) = \{\lambda_1 v_1 + \cdots + \lambda_m v_m : \lambda_1, \ldots, \lambda_m \in \mathbb{F}\}.$$

Definition 2.6 (Linear Independence). Vectors $v_1, \ldots, v_m \in V$ are *linearly independent* if the only solution to $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ is $\lambda_1 = \cdots = \lambda_m = 0$. They are *linearly dependent* otherwise.

Lemma 2.5 (Linear Dependence Lemma). *If $v_1, \ldots, v_m$ are linearly dependent and $v_1 \neq 0$, then there exists $j \in \{2, \ldots, m\}$ such that $v_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$ and*

$$\operatorname{span}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_m) = \operatorname{span}(v_1, \ldots, v_m).$$

Proof. Since the vectors are linearly dependent, there exist $\lambda_1, \ldots, \lambda_m$, not all zero, with $\sum_{i=1}^m \lambda_i v_i = 0$. Let j be the largest index with $\lambda_j \neq 0$. Then $j \geq 2$ (since $v_1 \neq 0$ means $\lambda_1 = 0$ would force all $\lambda_i = 0$ if $j = 1$). We have $v_j = -\lambda_j^{-1} \sum_{i=1}^{j-1} \lambda_i v_i \in \operatorname{span}(v_1, \ldots, v_{j-1})$. Any vector in $\operatorname{span}(v_1, \ldots, v_m)$ can replace its v_j-component using this expression, so removing v_j does not change the span. □

2.4 Basis and Dimension

Definition 2.7 (Basis). A *basis* of V is a list of vectors in V that is linearly independent and spans V.

Theorem 2.6 (Bases Have the Same Length). *If V is finite-dimensional, then any two bases of V have the same number of elements.*

Proof. Let $\{u_1, \ldots, u_m\}$ and $\{w_1, \ldots, w_n\}$ be two bases. We show $m \leq n$. Since the w's span V, the list $w_1, \ldots, w_n, u_1$ is linearly dependent. By the linear dependence lemma, we can remove some w_j while preserving the span. Repeating this process, each time we add a u_i and remove a w_j, after m steps we have a spanning list consisting of all m vectors u_i and $n - m$ remaining w's. This requires $n - m \geq 0$, so $m \leq n$. By symmetry, $n \leq m$, hence $m = n$. □

Definition 2.8 (Dimension)**.** The *dimension* of a finite-dimensional vector space V, denoted $\dim V$, is the number of elements in any basis of V.

Proposition 2.7 (Basis Extension)**.** *Every linearly independent list in a finite-dimensional vector space can be extended to a basis.*

Proof. Let $v_1, \ldots, v_m$ be linearly independent in V with $\dim V = n$. If $m = n$, we are done (a linearly independent list of length n in an n-dimensional space must span). If $m < n$, the list does not span V, so there exists $v_{m+1} \notin \operatorname{span}(v_1, \ldots, v_m)$. Then $v_1, \ldots, v_{m+1}$ is linearly independent. Repeat until the list has length n. □

Proposition 2.8 (Dimension of a Subspace)**.** *If U is a subspace of a finite-dimensional space V, then $\dim U \leq \dim V$. Equality holds if and only if $U = V$.*

Proof. Any linearly independent list in U is also linearly independent in V, so its length is at most $\dim V$. Hence $\dim U \leq \dim V$. If $\dim U = \dim V = n$, then a basis of U is a linearly independent list of length n in V, hence a basis of V (by Theorem 2.6 and the fact that n independent vectors in an n-dimensional space must span). Thus $U = V$. □

3 Linear Maps

Definition 3.1 (Linear Map)**.** A function $T : V \to W$ between vector spaces over $\mathbb{F}$ is a *linear map* if $T(u + v) = Tu + Tv$ and $T(\lambda v) = \lambda Tv$ for all $u, v \in V$ and $\lambda \in \mathbb{F}$.

Definition 3.2 (Null Space and Range)**.** For a linear map $T : V \to W$:

- The *null space* (kernel) is $\operatorname{null} T = \{v \in V : Tv = 0\}$.
- The *range* (image) is $\operatorname{range} T = \{Tv : v \in V\}$.

Both are subspaces (of V and W, respectively).

Proposition 3.1. *A linear map T is injective if and only if* $\operatorname{null} T = \{0\}$.

Proof. ($\Rightarrow$) If T is injective and $Tv = 0 = T0$, then $v = 0$. ($\Leftarrow$) If $\operatorname{null} T = \{0\}$ and $Tu = Tv$, then $T(u - v) = 0$, so $u - v = 0$, hence $u = v$. □

Theorem 3.2 (Rank-Nullity Theorem)**.** *If V is finite-dimensional and $T : V \to W$ is linear, then*

$$\dim V = \dim(\operatorname{null} T) + \dim(\operatorname{range} T).$$

Proof. Let $u_1, \ldots, u_m$ be a basis of $\operatorname{null} T$ and extend it to a basis $u_1, \ldots, u_m, v_1, \ldots, v_k$ of V. We claim $Tv_1, \ldots, Tv_k$ is a basis of $\operatorname{range} T$.

Spanning: Any $w \in \operatorname{range} T$ has $w = T(\lambda_1 u_1 + \cdots + \lambda_m u_m + \mu_1 v_1 + \cdots + \mu_k v_k) = \mu_1 Tv_1 + \cdots + \mu_k Tv_k$ (since $Tu_i = 0$).

Independence: If $\mu_1 Tv_1 + \cdots + \mu_k Tv_k = 0$, then $T(\mu_1 v_1 + \cdots + \mu_k v_k) = 0$, so $\mu_1 v_1 + \cdots + \mu_k v_k \in \operatorname{null} T$. Thus $\mu_1 v_1 + \cdots + \mu_k v_k = \alpha_1 u_1 + \cdots + \alpha_m u_m$ for some scalars. Since the full list is a basis, all coefficients are zero, so $\mu_i = 0$ for all i.

Therefore $\dim(\operatorname{range} T) = k$ and $\dim V = m + k = \dim(\operatorname{null} T) + \dim(\operatorname{range} T)$. □

Corollary 3.3. *Let $T : V \to W$ be linear with $\dim V = \dim W = n$. The following are equivalent:*

(i) T is injective.
(ii) T is surjective.
(iii) T is invertible (bijective).

Proof. By rank-nullity, $\dim(\operatorname{null} T) + \dim(\operatorname{range} T) = n$. Injectivity means $\dim(\operatorname{null} T) = 0$, giving $\dim(\operatorname{range} T) = n = \dim W$, which gives surjectivity (and vice versa). Invertibility is injectivity plus surjectivity. □

3.1 Matrix Representation

Fix bases $\{v_1, \ldots, v_n\}$ of V and $\{w_1, \ldots, w_m\}$ of W. Any linear map $T : V \to W$ is uniquely determined by its values on the basis: $Tv_j = \sum_{i=1}^{m} A_{ij} w_i$. The $m \times n$ matrix $A = (A_{ij})$ is the *matrix of* T with respect to the chosen bases. Composition of linear maps corresponds to matrix multiplication: if $S : U \to V$ has matrix B and $T : V \to W$ has matrix A, then $T \circ S$ has matrix AB.

Proposition 3.4 (Isomorphism of Finite-Dimensional Spaces)**.** *Two finite-dimensional vector spaces over $\mathbb{F}$ are isomorphic if and only if they have the same dimension.*

Proof. ($\Rightarrow$) If $T : V \to W$ is an isomorphism and $\{v_1, \ldots, v_n\}$ is a basis of V, then $\{Tv_1, \ldots, Tv_n\}$ is a basis of W (injectivity gives independence, surjectivity gives spanning), so $\dim W = n = \dim V$.

($\Leftarrow$) If $\dim V = \dim W = n$, pick bases $\{v_i\}$ of V and $\{w_i\}$ of W. Define $T : V \to W$ by $Tv_i = w_i$ and extend linearly. This is well-defined, linear, and bijective (it maps a basis to a basis). □

3.2 Change of Basis and Similarity

Definition 3.3 (Change-of-Basis Matrix)**.** Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ and $\mathcal{B}' = \{v'_1, \ldots, v'_n\}$ be two bases of V. The *change-of-basis matrix* from $\mathcal{B}$ to $\mathcal{B}'$ is the $n \times n$ matrix P whose j-th column is the coordinate vector of v_j in the basis $\mathcal{B}'$: $v_j = \sum_{i=1}^{n} P_{ij} v'_i$.

Proposition 3.5 (Similarity)**.** *If $T : V \to V$ has matrix A with respect to basis $\mathcal{B}$ and matrix A' with respect to basis $\mathcal{B}'$, then $A' = P^{-1}AP$, where P is the change-of-basis matrix from $\mathcal{B}'$ to $\mathcal{B}$.*

Proof. Let $[x]_{\mathcal{B}}$ denote the coordinate vector of x relative to $\mathcal{B}$. Then $P[x]_{\mathcal{B}'} = [x]_{\mathcal{B}}$ and $A[x]_{\mathcal{B}} = [Tx]_{\mathcal{B}}$. Thus $A'[x]_{\mathcal{B}'} = [Tx]_{\mathcal{B}'} = P^{-1}[Tx]_{\mathcal{B}} = P^{-1}A[x]_{\mathcal{B}} = P^{-1}AP[x]_{\mathcal{B}'}$. □

Bellman Equation as a Linear System

For a fixed policy π in a finite MDP, the Bellman equation $V^\pi = r^\pi + \gamma P^\pi V^\pi$ is a linear system $(I - \gamma P^\pi)V^\pi = r^\pi$. It has a unique solution because $I - \gamma P^\pi$ is invertible (its eigenvalues are $1 - \gamma\lambda_i$ where $|\lambda_i| \leq 1$ and $\gamma < 1$, so all eigenvalues are nonzero). Solving linear systems is one of the most fundamental operations in computational RL.

4 Determinants

We define the determinant for $n \times n$ matrices over $\mathbb{F}$ and establish the key properties.

Definition 4.1 (Determinant)**.** For $A \in \mathbb{F}^{n \times n}$, the *determinant* is defined recursively by cofactor expansion along the first row:

$$\det A = \sum_{j=1}^{n} (-1)^{1+j} A_{1j} \det A'_{1j},$$

where A'_{1j} is the $(n-1) \times (n-1)$ matrix obtained by deleting row 1 and column j. The base case is $\det(a) = a$ for 1×1 matrices.

Remark 4.1. Equivalently, the determinant can be expressed via the *Leibniz formula*: $\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} A_{i,\sigma(i)}$, where the sum runs over all permutations of $\{1, \ldots, n\}$ and $\operatorname{sgn}(\sigma)$ is the sign of the permutation. This formula makes it clear that the determinant is multilinear and alternating in the rows.

Theorem 4.1 (Properties of the Determinant)**.** *For $A, B \in \mathbb{F}^{n\times n}$:*

1. $\det I = 1$.
2. *The determinant is multilinear and alternating in the rows of A.*
3. *If A has two equal rows, then* $\det A = 0$.
4. $\det A^T = \det A$.
5. *Row operations: swapping two rows multiplies* det *by -1; adding a scalar multiple of one row to another leaves* det *unchanged.*
6. *Cofactor expansion works along* any *row or column, not just the first row.*

Proof. Properties (1)–(5) follow from the Leibniz formula and induction on n. We prove (3) and (5) explicitly.

(3) If rows p and q (with $p \neq q$) are equal, swapping them leaves A unchanged. By the alternating property (from multilinearity applied to the sum of two equal rows), $\det A = -\det A$, so $\det A = 0$.

(5) Adding c times row p to row q ($p \neq q$): by multilinearity in row q, $\det A' = \det A + c\cdot\det B$ where B is the matrix with row q replaced by row p. Since B has two equal rows, $\det B = 0$ by (3).

(6) Property (4) ($\det A^T = \det A$) shows that every row property also holds for columns. In particular, cofactor expansion along any column gives the same value. To see that expansion along row i also works: use row swaps to bring row i to the top (costing a factor of $(-1)^{i-1}$), expand along the first row, and verify the signs match. □

Theorem 4.2 (Multiplicativity)**.** *For $A, B \in \mathbb{F}^{n\times n}$,* $\det(AB) = \det A \cdot \det B$.

Proof. If A is not invertible, then $\operatorname{rank}(AB) \leq \operatorname{rank}(A) < n$, so AB is not invertible and both sides are zero. If A is invertible, it is a product of elementary matrices (row operations). For elementary matrices E, one verifies directly that $\det(EB) = \det(E)\det(B)$: a row swap gives $\det(E) = -1$, a row scaling by c gives $\det(E) = c$, and a row addition gives $\det(E) = 1$. The result follows by induction on the number of elementary factors. □

Corollary 4.3. *A is invertible if and only if* $\det A \neq 0$.

Proof. ($\Rightarrow$) If A is invertible, $1 = \det I = \det(AA^{-1}) = \det A \cdot \det A^{-1}$, so $\det A \neq 0$. ($\Leftarrow$) If A is not invertible, row reduction produces a row of zeros, giving $\det A = 0$. □

Corollary 4.4. *If A is invertible, then* $\det(A^{-1}) = (\det A)^{-1}$.

Proof. $\det A \cdot \det(A^{-1}) = \det(AA^{-1}) = \det I = 1$. □

5 Eigenvalues and Eigenvectors

Definition 5.1 (Eigenvalue, Eigenvector, Eigenspace)**.** Let $T : V \to V$ be a linear operator on a finite-dimensional space V. A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of T if there exists a nonzero $v \in V$ with $Tv = \lambda v$. Such a v is an *eigenvector* corresponding to λ. The *eigenspace* of λ is

$$E_\lambda = \operatorname{null}(T - \lambda I) = \{v \in V : Tv = \lambda v\}.$$

This is a subspace of V; its dimension is the *geometric multiplicity* of λ.

Definition 5.2 (Characteristic Polynomial)**.** For $A \in \mathbb{F}^{n\times n}$, the *characteristic polynomial* is $p_A(\lambda) = \det(\lambda I - A)$. This is a polynomial of degree n in λ.

Theorem 5.1. *λ is an eigenvalue of A if and only if* $\det(\lambda I - A) = 0$.

Proof. λ is an eigenvalue $\iff$ there exists $v \neq 0$ with $Av = \lambda v$ $\iff$ $(\lambda I - A)v = 0$ has a nontrivial solution $\iff$ $\lambda I - A$ is not invertible $\iff$ $\det(\lambda I - A) = 0$. □

Definition 5.3 (Algebraic Multiplicity)**.** The *algebraic multiplicity* of an eigenvalue λ_0 is its multiplicity as a root of the characteristic polynomial $p_A(\lambda)$. That is, if $p_A(\lambda) = (\lambda - \lambda_0)^k q(\lambda)$ where $q(\lambda_0) \neq 0$, then the algebraic multiplicity of λ_0 is k.

Theorem 5.2 (Geometric $\leq$ Algebraic Multiplicity)**.** *For any eigenvalue λ_0 of A,*

$$1 \leq \dim E_{\lambda_0} \leq \textit{algebraic multiplicity of } \lambda_0.$$

Proof. The lower bound holds by definition (eigenvalues have at least one eigenvector). For the upper bound, let $g = \dim E_{\lambda_0}$ and choose a basis $v_1, \ldots, v_g$ of E_{λ_0}. Extend to a basis $v_1, \ldots, v_g, v_{g+1}, \ldots, v_n$ of V. In this basis, the matrix of T has the form

$$B = \begin{pmatrix} \lambda_0 I_g & C \\ 0 & D \end{pmatrix},$$

where I_g is the $g \times g$ identity, C is $g \times (n-g)$, and D is $(n-g) \times (n-g)$. The block-triangular structure gives $p_A(\lambda) = \det(\lambda I - B) = (\lambda - \lambda_0)^g \det(\lambda I_{n-g} - D)$. Thus $(\lambda - \lambda_0)^g$ divides $p_A(\lambda)$, so the algebraic multiplicity is at least g. □

Theorem 5.3 (Eigenvectors for Distinct Eigenvalues)**.** *If $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T with corresponding eigenvectors $v_1, \ldots, v_m$, then $v_1, \ldots, v_m$ are linearly independent.*

Proof. By induction on m. The case $m = 1$ is clear ($v_1 \neq 0$). Suppose the result holds for $m-1$ eigenvectors and assume $\sum_{i=1}^m c_i v_i = 0$. Apply T to get $\sum_{i=1}^m c_i \lambda_i v_i = 0$. Subtract λ_m times the original equation: $\sum_{i=1}^{m-1} c_i(\lambda_i - \lambda_m) v_i = 0$. By the inductive hypothesis, $v_1, \ldots, v_{m-1}$ are linearly independent, so $c_i(\lambda_i - \lambda_m) = 0$ for $i = 1, \ldots, m-1$. Since the eigenvalues are distinct, $\lambda_i - \lambda_m \neq 0$, so $c_i = 0$ for $i < m$. The original equation then gives $c_m v_m = 0$, so $c_m = 0$. □

Corollary 5.4 (Sufficient Condition for Diagonalizability)**.** *If $A \in \mathbb{F}^{n \times n}$ has n distinct eigenvalues, then A is diagonalizable.*

Proof. By Theorem 5.3, the n eigenvectors are linearly independent and thus form a basis. If P is the matrix whose columns are these eigenvectors, then $P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. □

Theorem 5.5 (Full Diagonalizability Criterion)**.** *A matrix $A \in \mathbb{F}^{n \times n}$ is diagonalizable if and only if the geometric multiplicity equals the algebraic multiplicity for every eigenvalue.*

Proof. ($\Rightarrow$) If A is diagonalizable, then $A = P \operatorname{diag}(\lambda_1, \ldots, \lambda_n) P^{-1}$. For each distinct eigenvalue λ, the number of times it appears on the diagonal is both its algebraic multiplicity (from the characteristic polynomial of a diagonal matrix) and its geometric multiplicity (the corresponding columns of P form a basis of E_λ).

($\Leftarrow$) Suppose every eigenvalue has geometric multiplicity equal to its algebraic multiplicity. Let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues with algebraic multiplicities $m_1, \ldots, m_k$. Since $\sum m_i = n$ (the characteristic polynomial has degree n) and $\dim E_{\lambda_i} = m_i$, we need to show that the eigenspaces together span V.

Choose a basis for each E_{λ_i}. The union of these bases has $m_1 + \cdots + m_k = n$ vectors. We claim they are linearly independent. Suppose $\sum_{i=1}^k w_i = 0$ where $w_i \in E_{\lambda_i}$. Since eigenvectors from distinct eigenspaces are linearly independent (by applying Theorem 5.3 to any nonzero members), the only solution is $w_i = 0$ for all i. Within each eigenspace, the chosen basis vectors are independent by construction. Therefore the full collection of n vectors is linearly independent and spans V. □

Not Every Matrix Is Diagonalizable

The matrix $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has characteristic polynomial λ^2, so $\lambda = 0$ has algebraic multiplicity 2. But $E_0 = \operatorname{span}\{e_1\}$ has dimension 1. Since $1 < 2$, the matrix is not diagonalizable.

5.1 The Cayley–Hamilton Theorem

Theorem 5.6 (Cayley–Hamilton). *Every matrix satisfies its own characteristic polynomial: if $p_A(\lambda) = \det(\lambda I - A)$, then $p_A(A) = 0$ (the zero matrix).*

Proof. Write $p_A(\lambda) = \det(\lambda I - A) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0$. The adjugate (classical adjoint) matrix $\operatorname{adj}(\lambda I - A)$ is a matrix whose entries are polynomials in λ of degree at most $n-1$, so we can write

$$\operatorname{adj}(\lambda I - A) = B_{n-1}\lambda^{n-1} + B_{n-2}\lambda^{n-2} + \cdots + B_0,$$

where $B_0, \ldots, B_{n-1}$ are constant matrices. By the identity $M \cdot \operatorname{adj}(M) = (\det M)I$ applied to $M = \lambda I - A$:

$$(\lambda I - A)(B_{n-1}\lambda^{n-1} + \cdots + B_0) = (\lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0)I.$$

Expanding the left side and matching powers of λ:

$$\begin{aligned}
\lambda^n &: \quad B_{n-1} = I, \\
\lambda^{n-1} &: \quad B_{n-2} - AB_{n-1} = c_{n-1}I, \\
\lambda^{n-2} &: \quad B_{n-3} - AB_{n-2} = c_{n-2}I, \\
&\vdots \\
\lambda^1 &: \quad B_0 - AB_1 = c_1 I, \\
\lambda^0 &: \quad -AB_0 = c_0 I.
\end{aligned}$$

Multiply the equation for λ^k on the left by A^k and sum:

$$A^n B_{n-1} + A^{n-1}(B_{n-2} - AB_{n-1}) + \cdots + A(B_0 - AB_1) + (-AB_0) = A^n + c_{n-1}A^{n-1} + \cdots + c_0 I.$$

The left side telescopes to 0 (each $A^k B_{k-1}$ appears with $+$ from one term and $-$ from the next). The right side is $p_A(A)$. Therefore $p_A(A) = 0$. □

Remark 5.1. The Cayley–Hamilton theorem implies that A^{-1} (when it exists) can be expressed as a polynomial in A: since $p_A(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I = 0$ and $c_0 = (-1)^n \det A \neq 0$, we get $A^{-1} = -c_0^{-1}(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I)$.

Eigenvalues and Convergence Rates

Eigenvalue analysis is central to understanding convergence rates. The spectral radius of a transition matrix P determines the mixing time of a Markov chain. The condition number $\kappa = \lambda_{\max}/\lambda_{\min}$ of the Hessian determines gradient descent convergence speed: gradient descent on a quadratic with condition number κ requires $O(\kappa \log(1/\varepsilon))$ iterations. The Cayley–Hamilton theorem is used to analyze matrix polynomials that appear in multi-step TD learning algorithms.

6 Inner Product Spaces

Definition 6.1 (Inner Product)**.** An *inner product* on a real vector space V is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying:

1. **Symmetry:** $\langle u, v \rangle = \langle v, u \rangle$.
2. **Linearity in the first argument:** $\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle$.
3. **Positive definiteness:** $\langle v, v \rangle \geq 0$ with equality iff $v = 0$.

The pair $(V, \langle \cdot, \cdot \rangle)$ is an *inner product space*. The *norm* induced by the inner product is $\|v\| = \sqrt{\langle v, v \rangle}$.

Remark 6.1. For complex vector spaces, condition (1) becomes *conjugate symmetry*: $\langle u, v \rangle = \overline{\langle v, u \rangle}$, and condition (2) holds in the first argument (making the inner product conjugate-linear in the second). For this document we work primarily over $\mathbb{R}$; Phase 01 will use complex inner products when needed.

Remark 6.2. The standard inner product on $\mathbb{R}^n$ is $\langle x, y \rangle = \sum_{i=1}^n x_i y_i = x^T y$.

Theorem 6.1 (Cauchy–Schwarz Inequality)**.** *For all u, v in an inner product space,*

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|,$$

where $\|w\| = \sqrt{\langle w, w \rangle}$. Equality holds if and only if u and v are linearly dependent.

Proof. If $v = 0$, both sides are zero. Assume $v \neq 0$ and let $t \in \mathbb{R}$. Then

$$0 \leq \|u - tv\|^2 = \langle u - tv, u - tv \rangle = \|u\|^2 - 2t\langle u, v \rangle + t^2 \|v\|^2.$$

This is a nonnegative quadratic in t, so its discriminant is non-positive: $4\langle u, v \rangle^2 - 4\|u\|^2\|v\|^2 \leq 0$, giving $|\langle u, v \rangle| \leq \|u\|\|v\|$.

Equality holds iff the discriminant is zero, i.e., the minimum of the quadratic is zero, meaning $u - tv = 0$ for some t, i.e., u and v are linearly dependent. □

Corollary 6.2 (Triangle Inequality)**.** *For all u, v in an inner product space, $\|u+v\| \leq \|u\|+\|v\|$.*

Proof. $\|u + v\|^2 = \|u\|^2 + 2\langle u, v \rangle + \|v\|^2 \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2$, using Cauchy–Schwarz. Taking square roots gives the result. □

6.1 Orthogonality and Gram–Schmidt

Definition 6.2 (Orthogonal, Orthonormal)**.** Vectors u, v are *orthogonal* (written $u \perp v$) if $\langle u, v \rangle = 0$. A list $e_1, \ldots, e_m$ is *orthonormal* if $\langle e_i, e_j \rangle = \delta_{ij}$ (the Kronecker delta).

Proposition 6.3. *An orthonormal list is linearly independent.*

Proof. If $\sum c_i e_i = 0$, take the inner product with e_j: $0 = \langle \sum c_i e_i, e_j \rangle = \sum c_i \langle e_i, e_j \rangle = c_j$. □

Theorem 6.4 (Pythagorean Theorem)**.** *If $e_1, \ldots, e_m$ are pairwise orthogonal, then $\|e_1 + \cdots + e_m\|^2 = \|e_1\|^2 + \cdots + \|e_m\|^2$.*

Proof. $\|e_1 + \cdots + e_m\|^2 = \langle \sum_i e_i, \sum_j e_j \rangle = \sum_{i,j} \langle e_i, e_j \rangle = \sum_i \|e_i\|^2$, since the cross terms vanish by orthogonality. □

Theorem 6.5 (Gram–Schmidt Process)**.** *Given a linearly independent list $v_1, \ldots, v_m$ in an inner product space, there exists an orthonormal list $e_1, \ldots, e_m$ such that* $\operatorname{span}(e_1, \ldots, e_k) = \operatorname{span}(v_1, \ldots, v_k)$ *for each $k = 1, \ldots, m$.*

Proof. **Algorithm:** Set $e_1 = v_1/\|v_1\|$. For $k = 2, \ldots, m$, define

$$w_k = v_k - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j, \qquad e_k = \frac{w_k}{\|w_k\|}.$$

Well-defined: $w_k \neq 0$ because $v_k \notin \operatorname{span}(v_1, \ldots, v_{k-1}) = \operatorname{span}(e_1, \ldots, e_{k-1})$ by linear independence.

Orthogonality: For $j < k$, $\langle w_k, e_j \rangle = \langle v_k, e_j \rangle - \langle v_k, e_j \rangle = 0$.

Span preservation: By construction, $e_k \in \operatorname{span}(v_1, \ldots, v_k)$ and $v_k = w_k + \sum \langle v_k, e_j \rangle e_j \in \operatorname{span}(e_1, \ldots, e_k)$, so the spans agree at each step. □

Corollary 6.6. *Every finite-dimensional inner product space has an orthonormal basis.*

Proof. Apply Gram–Schmidt to any basis. □

6.2 Orthogonal Complement and Decomposition

Definition 6.3 (Orthogonal Complement)**.** For a subset S of an inner product space V, the *orthogonal complement* is

$$S^\perp = \{v \in V : \langle v, s \rangle = 0 \text{ for all } s \in S\}.$$

Proposition 6.7. *$S^\perp$ is a subspace of V for any subset S.*

Proof. $0 \in S^\perp$ (since $\langle 0, s \rangle = 0$). If $u, v \in S^\perp$ and $\alpha \in \mathbb{F}$, then $\langle u + v, s \rangle = \langle u, s \rangle + \langle v, s \rangle = 0$ and $\langle \alpha u, s \rangle = \alpha \langle u, s \rangle = 0$ for all $s \in S$. □

Theorem 6.8 (Orthogonal Decomposition)**.** *If U is a finite-dimensional subspace of an inner product space V, then $V = U \oplus U^\perp$.*

Proof. $U \cap U^\perp = \{0\}$**:** If $v \in U \cap U^\perp$, then $\langle v, v \rangle = 0$, so $v = 0$ by positive definiteness.

$V = U + U^\perp$**:** Let $v \in V$ and let $e_1, \ldots, e_m$ be an orthonormal basis of U. Set $u = \sum_{i=1}^m \langle v, e_i \rangle e_i \in U$ and $w = v - u$. For each e_j: $\langle w, e_j \rangle = \langle v, e_j \rangle - \langle v, e_j \rangle = 0$, so $w \in U^\perp$. Thus $v = u + w \in U + U^\perp$.

By Proposition 2.3, $V = U \oplus U^\perp$. □

Corollary 6.9 (Dimension Formula)**.** *If U is a subspace of a finite-dimensional inner product space V, then $\dim U + \dim U^\perp = \dim V$ and $(U^\perp)^\perp = U$.*

Proof. $\dim V = \dim U + \dim U^\perp$ follows from the direct sum decomposition. For the second claim: $U \subseteq (U^\perp)^\perp$ (if $u \in U$, then $\langle u, w \rangle = 0$ for all $w \in U^\perp$). Since $\dim (U^\perp)^\perp = \dim V - \dim U^\perp = \dim U$, and $U \subseteq (U^\perp)^\perp$ with equal dimensions, $U = (U^\perp)^\perp$. □

6.3 Orthogonal Projection and Best Approximation

Theorem 6.10 (Best Approximation)**.** *Let U be a finite-dimensional subspace of an inner product space V and let $v \in V$. There exists a unique $u_0 \in U$ minimizing $\|v - u\|$ over $u \in U$, given by the* orthogonal projection*:*

$$u_0 = \operatorname{proj}_U(v) = \sum_{i=1}^m \langle v, e_i \rangle e_i,$$

where $e_1, \ldots, e_m$ is any orthonormal basis of U. Moreover, $v - u_0 \perp U$.

Proof. Define $u_0 = \sum \langle v, e_i \rangle e_i$. For any e_j, $\langle v - u_0, e_j \rangle = \langle v, e_j \rangle - \langle v, e_j \rangle = 0$, so $v - u_0 \perp U$.

For any $u \in U$, write $u - u_0 \in U$ and note $(v - u_0) \perp (u_0 - u)$:

$$\|v - u\|^2 = \|(v - u_0) + (u_0 - u)\|^2 = \|v - u_0\|^2 + \|u_0 - u\|^2 \geq \|v - u_0\|^2,$$

with equality iff $u = u_0$, by the Pythagorean theorem (Theorem 6.4). □

Proposition 6.11 (Bessel's Inequality)**.** *Let $e_1, \ldots, e_m$ be an orthonormal list in an inner product space V. For any $v \in V$,*

$$\sum_{i=1}^{m} |\langle v, e_i \rangle|^2 \leq \|v\|^2.$$

Equality holds if and only if $v \in \operatorname{span}(e_1, \ldots, e_m)$.

Proof. Let $U = \operatorname{span}(e_1, \ldots, e_m)$ and $u_0 = \operatorname{proj}_U(v) = \sum \langle v, e_i \rangle e_i$. By the Pythagorean theorem, $\|v\|^2 = \|u_0\|^2 + \|v - u_0\|^2$. Since the e_i are orthonormal, $\|u_0\|^2 = \sum |\langle v, e_i \rangle|^2$. Hence $\sum |\langle v, e_i \rangle|^2 = \|v\|^2 - \|v - u_0\|^2 \leq \|v\|^2$. Equality holds iff $v - u_0 = 0$, i.e., $v = u_0 \in U$. □

Orthogonal Projection: The Best Approximation

The orthogonal projection $\operatorname{proj}_U(v)$ is the unique element of U closest to v. This is the geometric essence of least squares: given data and a model subspace, the least squares solution is the orthogonal projection of the data onto the model space. This principle generalizes to Hilbert spaces in functional analysis (Phase 01).

7 The Spectral Theorem

We now restrict to real symmetric matrices, which are the most important case for optimization and machine learning.

Theorem 7.1 (Real Eigenvalues of Symmetric Matrices)**.** *Every real symmetric matrix has only real eigenvalues.*

Proof. Let $A \in \mathbb{R}^{n \times n}$ be symmetric with $Av = \lambda v$ for some $v \in \mathbb{C}^n$, $v \neq 0$. Consider $\bar{v}^T A v$. On one hand, $\bar{v}^T A v = \bar{v}^T (\lambda v) = \lambda \bar{v}^T v = \lambda \sum |v_i|^2$. On the other hand, since $A = A^T$ is real, $\bar{v}^T A v = (A^T \bar{v})^T v = (A\bar{v})^T v = (\bar{\lambda}\bar{v})^T v = \bar{\lambda}\bar{v}^T v = \bar{\lambda} \sum |v_i|^2$. Since $\sum |v_i|^2 > 0$, we get $\lambda = \bar{\lambda}$, so $\lambda \in \mathbb{R}$. □

Theorem 7.2 (Orthogonality of Eigenvectors)**.** *Eigenvectors of a real symmetric matrix corresponding to distinct eigenvalues are orthogonal.*

Proof. Let $Av = \lambda v$ and $Aw = \mu w$ with $\lambda \neq \mu$. Then $\lambda \langle v, w \rangle = \langle Av, w \rangle = \langle v, A^T w \rangle = \langle v, Aw \rangle = \mu \langle v, w \rangle$. So $(\lambda - \mu)\langle v, w \rangle = 0$. Since $\lambda \neq \mu$, $\langle v, w \rangle = 0$. □

Theorem 7.3 (Real Spectral Theorem)**.** *Every real symmetric matrix $A \in \mathbb{R}^{n \times n}$ is orthogonally diagonalizable: there exists an orthogonal matrix Q (i.e., $Q^T Q = I$) and a diagonal matrix $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ such that $A = Q \Lambda Q^T$.*

Proof. We proceed by induction on n. The case $n = 1$ is trivial.

Assume the result holds for $(n-1) \times (n-1)$ symmetric matrices. By Theorem 7.1, A has a real eigenvalue λ_1 with a unit eigenvector $v_1 \in \mathbb{R}^n$. Extend v_1 to an orthonormal basis $\{v_1, v_2, \ldots, v_n\}$ of $\mathbb{R}^n$ (using Gram–Schmidt on any extension). Let $P = [v_1 \mid v_2 \mid \cdots \mid v_n]$. Then

$$P^T A P = \begin{pmatrix} \lambda_1 & b^T \\ b & B \end{pmatrix},$$

where $b \in \mathbb{R}^{n-1}$ and $B \in \mathbb{R}^{(n-1)\times(n-1)}$. Since P^TAP is symmetric (as A is), the block structure gives $b^T = b^T$ (automatically) and we need $b = 0$. Indeed, the first column of P^TAP is $P^TAv_1 = P^T(\lambda_1 v_1) = \lambda_1 e_1$, so $b = 0$. Also B is symmetric.

By the inductive hypothesis, $B = Q'\Lambda'(Q')^T$ for some orthogonal Q' and diagonal Λ'. Set $Q_0 = \begin{pmatrix} 1 & 0 \\ 0 & Q' \end{pmatrix}$ and $Q = PQ_0$. Then Q is orthogonal and $Q^TAQ = Q_0^T(P^TAP)Q_0 = \operatorname{diag}(\lambda_1, \Lambda') = \Lambda$. □

Corollary 7.4 (Spectral Decomposition). *If $A = Q\Lambda Q^T$ with $Q = [q_1 \mid \cdots \mid q_n]$, then $A = \sum_{i=1}^n \lambda_i q_i q_i^T$. Each $q_i q_i^T$ is the rank-1 orthogonal projection onto* $\operatorname{span}\{q_i\}$.

Proof. $A = Q\Lambda Q^T = \sum_{i=1}^n \lambda_i q_i q_i^T$ follows from the matrix product $Q\Lambda = [\lambda_1 q_1 \mid \cdots \mid \lambda_n q_n]$. □

7.1 Positive Definiteness

Definition 7.1 (Positive Definite and Positive Semidefinite). A symmetric matrix A is:

- *Positive definite* ($A \succ 0$) if $x^TAx > 0$ for all $x \neq 0$.
- *Positive semidefinite* ($A \succeq 0$) if $x^TAx \geq 0$ for all x.

Theorem 7.5 (Characterizations of Positive Definiteness). *For a real symmetric matrix A, the following are equivalent:*

(i) A is positive definite.
(ii) All eigenvalues of A are positive.
(iii) $A = B^TB$ for some invertible matrix B.
(iv) All leading principal minors of A are positive (Sylvester's criterion).

For positive semidefiniteness, replace "positive" with "nonnegative" in (i) and (ii), "invertible" with "any" in (iii), and "positive" with "nonnegative" in (iv).

Proof. **(i) ⇔ (ii):** By the spectral theorem, $A = Q\Lambda Q^T$. For any $x \neq 0$, let $y = Q^Tx \neq 0$. Then $x^TAx = y^T\Lambda y = \sum \lambda_i y_i^2$. This is positive for all $y \neq 0$ iff all $\lambda_i > 0$.

(ii) ⇒ (iii): If all eigenvalues are positive, write $A = Q\Lambda Q^T = Q\Lambda^{1/2}\Lambda^{1/2}Q^T = (Q\Lambda^{1/2})^T(Q\Lambda^{1/2})$ where $\Lambda^{1/2} = \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})$. Set $B = Q\Lambda^{1/2}$; this is invertible since Q and $\Lambda^{1/2}$ are.

(iii) ⇒ (i): If $A = B^TB$ with B invertible, then for $x \neq 0$: $x^TAx = x^TB^TBx = \|Bx\|^2 > 0$ (since $Bx \neq 0$ by invertibility of B).

(i) ⇒ (iv): If $A \succ 0$ and A_k is the $k \times k$ leading principal submatrix, then for any $y \in \mathbb{R}^k$ with $y \neq 0$, let $x = (y, 0, \ldots, 0)^T$. Then $x^TAx = y^TA_ky > 0$, so $A_k \succ 0$, hence $\det A_k > 0$.

(iv) ⇒ (i): By induction on n. The case $n = 1$ is immediate ($A = (a_{11})$ with $a_{11} > 0$). For the inductive step, $\det A_{n-1} > 0$ gives $A_{n-1} \succ 0$ by the inductive hypothesis. One can then show via the Schur complement that $\det A > 0$ together with $A_{n-1} \succ 0$ implies $A \succ 0$. □

Proposition 7.6 (Cholesky Factorization). *A symmetric matrix A is positive definite if and only if there exists a unique lower triangular matrix L with positive diagonal entries such that $A = LL^T$.*

Proof sketch. **Existence:** Proceed by induction on n. For $n = 1$, $A = (a)$ with $a > 0$, so $L = (\sqrt{a})$. For the inductive step, partition

$$A = \begin{pmatrix} A_{n-1} & b \\ b^T & c \end{pmatrix}.$$

Since $A_{n-1} \succ 0$, by induction $A_{n-1} = L_0L_0^T$. Set $\ell = L_0^{-1}b$ (solving $L_0\ell = b$) and $\alpha = \sqrt{c - \|\ell\|^2}$ (which is real and positive since $\det A > 0$ implies $c - b^TA_{n-1}^{-1}b > 0$). Then $L = \left(\begin{smallmatrix} L_0 & 0 \\ \ell^T & \alpha \end{smallmatrix}\right)$ satisfies $A = LL^T$.

Uniqueness: If $A = LL^T = MM^T$, then $L^{-1}M(L^{-1}M)^T = I$, so $L^{-1}M$ is orthogonal. But $L^{-1}M$ is also lower triangular (product of lower triangular matrices). An orthogonal lower

triangular matrix must be diagonal with ± 1 entries. Since both L and M have positive diagonals, $L^{-1}M = I$. $\square$

7.2 The Rayleigh Quotient and Courant–Fischer

Definition 7.2 (Rayleigh Quotient)**.** For a symmetric matrix $A \in \mathbb{R}^{n\times n}$ and nonzero $x \in \mathbb{R}^n$, the *Rayleigh quotient* is

$$R_A(x) = \frac{x^T A x}{x^T x}.$$

Proposition 7.7 (Rayleigh Quotient and Eigenvalues)**.** *Let A be a real symmetric $n \times n$ matrix with eigenvalues $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$. Then:*

(i) $\lambda_1 \le R_A(x) \le \lambda_n$ *for all* $x \neq 0$.

(ii) $\lambda_n = \max_{x\neq 0} R_A(x)$ *and* $\lambda_1 = \min_{x\neq 0} R_A(x)$.

(iii) The maximum is attained at any eigenvector for λ_n; the minimum at any eigenvector for λ_1.

Proof. Write $A = Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)Q^T$ and let $y = Q^T x$, so $\|y\| = \|x\|$ and $x^T A x = \sum \lambda_i y_i^2$. Then

$$R_A(x) = \frac{\sum_i \lambda_i y_i^2}{\sum_i y_i^2}.$$

This is a weighted average of the eigenvalues with nonnegative weights $y_i^2/\|y\|^2$. Therefore $\lambda_1 \le R_A(x) \le \lambda_n$. Taking $x = q_n$ (the eigenvector for λ_n) gives $y = e_n$ and $R_A(x) = \lambda_n$. $\square$

Theorem 7.8 (Courant–Fischer Minimax Theorem)**.** *Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$. Then*

$$\lambda_k = \min_{\substack{S\subseteq\mathbb{R}^n \\ \dim S = k}} \max_{x\in S\setminus\{0\}} R_A(x) = \max_{\substack{S\subseteq\mathbb{R}^n \\ \dim S = n-k+1}} \min_{x\in S\setminus\{0\}} R_A(x).$$

Proof. We prove the first (min-max) characterization. Let $q_1, \ldots, q_n$ be orthonormal eigenvectors with $Aq_i = \lambda_i q_i$.

Upper bound: Take $S_0 = \mathrm{span}(q_1, \ldots, q_k)$ (dimension k). Any $x \in S_0$ has $x = \sum_{i=1}^k c_i q_i$, so $R_A(x) = \sum_{i=1}^k \lambda_i c_i^2 / \sum_{i=1}^k c_i^2 \le \lambda_k$. Hence $\max_{x\in S_0\setminus\{0\}} R_A(x) \le \lambda_k$, giving $\min_S \max_x R_A(x) \le \lambda_k$.

Lower bound: Let S be any k-dimensional subspace. The subspace $T = \mathrm{span}(q_k, q_{k+1}, \ldots, q_n)$ has dimension $n-k+1$. Since $\dim S + \dim T = k + (n-k+1) = n+1 > n$, the intersection $S \cap T$ is nontrivial. Pick a unit vector $x \in S \cap T$, so $x = \sum_{i=k}^n c_i q_i$. Then $R_A(x) = \sum_{i=k}^n \lambda_i c_i^2 \ge \lambda_k$. Hence $\max_{x\in S\setminus\{0\}} R_A(x) \ge \lambda_k$ for every k-dimensional S, giving $\min_S \max_x R_A(x) \ge \lambda_k$. $\square$

The Spectral Theorem in Practice

Every real symmetric matrix A can be written as $A = Q\Lambda Q^T$ where Q is orthogonal and Λ is diagonal. This decomposition is the foundation of:

1. **PCA** — the principal components are eigenvectors of the covariance matrix.
2. **Convexity** — a twice-differentiable function is convex iff its Hessian is positive semidefinite (all eigenvalues ≥ 0).
3. **Natural policy gradient** — the update direction is $F^{-1}\nabla J$, where F is the (symmetric, positive semidefinite) Fisher information matrix.
4. **Convergence analysis** — the Courant–Fischer theorem gives tight bounds on eigenvalues, which control convergence rates of iterative methods.

8 Quadratic Forms

Definition 8.1 (Quadratic Form)**.** A *quadratic form* on $\mathbb{R}^n$ is a function $Q : \mathbb{R}^n \to \mathbb{R}$ of the form $Q(x) = x^T Ax$ where $A \in \mathbb{R}^{n\times n}$ is symmetric.

Remark 8.1. Any expression $x^T Bx$ with B not necessarily symmetric can be rewritten as $x^T \left(\frac{B+B^T}{2}\right) x$, since $x^T Bx = x^T B^T x$ (both are scalars). So there is no loss of generality in assuming A symmetric.

Proposition 8.1 (Classification by Eigenvalues)**.** *Let $Q(x) = x^T Ax$ with A symmetric and eigenvalues $\lambda_1 \le \cdots \le \lambda_n$.*

(i) $Q(x) > 0$ for all $x \neq 0$ (positive definite) $\iff$ all $\lambda_i > 0$.
(ii) $Q(x) \ge 0$ for all x (positive semidefinite) $\iff$ all $\lambda_i \ge 0$.
(iii) Q takes both positive and negative values (indefinite) $\iff$ A has both positive and negative eigenvalues.

Proof. Immediate from the spectral theorem: $Q(x) = x^T Q_e \Lambda Q_e^T x = \sum \lambda_i y_i^2$ where $y = Q_e^T x$. □

Proposition 8.2 (Quadratic Approximation)**.** *If $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable, then near a critical point x_0 (where $\nabla f(x_0) = 0$):*

$$f(x_0 + h) = f(x_0) + \frac{1}{2} h^T Hh + o(\|h\|^2),$$

where $H = \nabla^2 f(x_0)$ is the Hessian matrix (which is symmetric). The nature of the critical point is determined by the quadratic form $h \mapsto h^T Hh$:

- *$H \succ 0 \Rightarrow x_0$ is a strict local minimum.*
- *$H \prec 0 \Rightarrow x_0$ is a strict local maximum.*
- *H indefinite $\Rightarrow x_0$ is a saddle point.*

Proof. The Taylor expansion is $f(x_0 + h) = f(x_0) + \nabla f(x_0)^T h + \frac{1}{2} h^T Hh + o(\|h\|^2)$. Since $\nabla f(x_0) = 0$, the linear term vanishes.

If $H \succ 0$, then $h^T Hh \ge \lambda_{\min}(H)\|h\|^2 > 0$ for $h \neq 0$, so for $\|h\|$ sufficiently small the $o(\|h\|^2)$ remainder is dominated by the quadratic term, giving $f(x_0 + h) > f(x_0)$.

If H is indefinite, there exist directions h_+ and h_- with $h_+^T Hh_+ > 0$ and $h_-^T Hh_- < 0$. Along these directions, f increases and decreases respectively, so x_0 is a saddle point. □

Quadratic Forms in Optimization and RL

Quadratic forms appear throughout optimization and RL. The second-order Taylor expansion gives $f(x + h) \approx f(x) + \nabla f(x)^T h + \frac{1}{2} h^T \nabla^2 f(x) h$. Newton's method uses this approximation: the step $h = -(\nabla^2 f)^{-1} \nabla f$ minimizes the quadratic model. In natural policy gradient methods, the Fisher information matrix plays the role of the Hessian, and the update direction is $h = -F^{-1} \nabla J(\theta)$.

9 Singular Value Decomposition

The SVD extends diagonalization to rectangular matrices.

Theorem 9.1 (Singular Value Decomposition)**.** *Every matrix $A \in \mathbb{R}^{m\times n}$ can be written as*

$$A = U\Sigma V^T,$$

where $U \in \mathbb{R}^{m\times m}$ is orthogonal, $V \in \mathbb{R}^{n\times n}$ is orthogonal, and $\Sigma \in \mathbb{R}^{m\times n}$ is "diagonal" with nonnegative entries $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_{\min(m,n)} \ge 0$ (the singular values*).*

Proof. The matrix $A^TA \in \mathbb{R}^{n\times n}$ is symmetric and positive semidefinite (since $x^TA^TAx = \|Ax\|^2 \geq 0$). By the spectral theorem, $A^TA = VDV^T$ where $D = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)$ with $\sigma_i \geq 0$ and V orthogonal.

For each $\sigma_i > 0$, define $u_i = Av_i/\sigma_i$. Then the u_i are orthonormal: $\langle u_i, u_j\rangle = \frac{1}{\sigma_i\sigma_j} v_i^TA^TAv_j = \frac{\sigma_j^2}{\sigma_i\sigma_j} v_i^Tv_j = \delta_{ij}$.

Extend $\{u_1, \ldots, u_r\}$ (where $r = \operatorname{rank} A$) to an orthonormal basis of $\mathbb{R}^m$. Forming $U = [u_1 \mid \cdots \mid u_m]$ gives $AV = U\Sigma$, hence $A = U\Sigma V^T$. □

Remark 9.1 (Geometric Interpretation). The SVD says: any linear map $\mathbb{R}^n \to \mathbb{R}^m$ can be decomposed as (1) rotate in the domain (apply V^T), (2) scale along coordinate axes (apply Σ), (3) rotate in the codomain (apply U). The singular values are the semi-axis lengths of the image of the unit ball.

Proposition 9.2 (Rank from SVD)**.** *The rank of A equals the number of nonzero singular values.*

Proof. $\operatorname{rank} A = \operatorname{rank}(U\Sigma V^T) = \operatorname{rank}\Sigma$ (since U and V are invertible), and $\operatorname{rank}\Sigma$ equals the number of nonzero diagonal entries. □

Theorem 9.3 (Eckart–Young: Best Low-Rank Approximation)**.** *Let $A = U\Sigma V^T$ be the SVD of A with singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$. For any $k < r$, the matrix $A_k = \sum_{i=1}^k \sigma_i u_i v_i^T$ minimizes $\|A - B\|_F$ over all matrices B of rank at most k, and $\|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \cdots + \sigma_r^2}$.*

Proof. Write $A = \sum_{i=1}^r \sigma_i u_i v_i^T$. Since the $u_iv_i^T$ have orthonormal columns and rows, $\|A\|_F^2 = \sum \sigma_i^2$ (the Frobenius norm depends only on singular values). For $A_k = \sum_{i=1}^k \sigma_i u_i v_i^T$, $\|A - A_k\|_F^2 = \sum_{i=k+1}^r \sigma_i^2$.

To show this is minimal, let B have rank $\leq k$, so $\operatorname{null} B$ has dimension $\geq n - k$. The space $\operatorname{span}(v_1, \ldots, v_{k+1})$ has dimension $k + 1$, so it intersects $\operatorname{null} B$ nontrivially. Pick a unit z in the intersection. Then $\|A - B\|_F^2 \geq \|(A - B)z\|^2 = \|Az\|^2$ (since $Bz = 0$). Now $z = \sum_{i=1}^{k+1} c_i v_i$ with $\sum c_i^2 = 1$, so $Az = \sum_{i=1}^{k+1} c_i\sigma_i u_i$ and $\|Az\|^2 = \sum_{i=1}^{k+1} c_i^2\sigma_i^2 \geq \sigma_{k+1}^2$. Repeating this argument with appropriate subspaces yields $\|A - B\|_F^2 \geq \sum_{i=k+1}^r \sigma_i^2$. □

9.1 The Pseudoinverse

Definition 9.1 (Moore–Penrose Pseudoinverse)**.** Let $A = U\Sigma V^T$ be the SVD with r nonzero singular values. The *Moore–Penrose pseudoinverse* is $A^+ = V\Sigma^+U^T$, where Σ^+ is obtained from Σ by transposing and replacing each nonzero σ_i with σ_i^{-1}.

Proposition 9.4 (Pseudoinverse and Least Squares)**.** *For any $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, the vector $x^* = A^+b$ minimizes $\|Ax - b\|^2$ over all $x \in \mathbb{R}^n$. Among all minimizers, x^* has the smallest norm.*

Proof. Write $A = U\Sigma V^T$, so $\|Ax - b\|^2 = \|U\Sigma V^Tx - b\|^2 = \|\Sigma y - c\|^2$ where $y = V^Tx$ and $c = U^Tb$ (using $\|Uw\| = \|w\|$). This splits into two parts: for $i \leq r$, the term $(\sigma_iy_i - c_i)^2$ is minimized by $y_i = c_i/\sigma_i$; for $i > r$, we get c_i^2 regardless of y_i. The minimum-norm solution sets $y_i = 0$ for $i > r$. Transforming back: $x^* = Vy^* = V\Sigma^+U^Tb = A^+b$. □

SVD in Machine Learning

SVD is the workhorse of dimensionality reduction and matrix approximation. In RL, low-rank approximations appear in linear function approximation (projecting value functions onto a low-dimensional feature space) and in compressing large state-action value tables. PCA is a direct application: the principal components are the right singular vectors of

the centered data matrix. The pseudoinverse provides the minimum-norm least-squares solution used in linear regression and temporal-difference learning with linear function approximation.

10 Trace, Matrix Norms, and the Matrix Exponential

10.1 The Trace

Definition 10.1 (Trace)**.** The *trace* of a square matrix $A \in \mathbb{F}^{n\times n}$ is $\operatorname{tr}(A) = \sum_{i=1}^{n} A_{ii}$.

Theorem 10.1 (Properties of the Trace)**.** *(i)* ***Linearity:*** $\operatorname{tr}(\alpha A + \beta B) = \alpha \operatorname{tr}(A) + \beta \operatorname{tr}(B)$.

(ii) ***Cyclic property:*** $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ *for* $A \in \mathbb{F}^{m\times n}$, $B \in \mathbb{F}^{n\times m}$.

(iii) ***Similarity invariance:*** *If* $B = P^{-1}AP$, *then* $\operatorname{tr}(B) = \operatorname{tr}(A)$.

(iv) ***Eigenvalue sum:*** $\operatorname{tr}(A) = \sum_{i=1}^{n} \lambda_i$ *where the* λ_i *are the eigenvalues of* A *(counted with algebraic multiplicity).*

(v) ***Transpose:*** $\operatorname{tr}(A^T) = \operatorname{tr}(A)$.

(vi) ***Inner product:*** $\operatorname{tr}(A^T B) = \sum_{i,j} A_{ij}B_{ij}$ *(the Frobenius inner product of matrices).*

Proof. **(i)** Immediate from the definition.

(ii) $\operatorname{tr}(AB) = \sum_{i=1}^{m}(AB)_{ii} = \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}B_{ji} = \sum_{j=1}^{n}\sum_{i=1}^{m} B_{ji}A_{ij} = \sum_{j=1}^{n}(BA)_{jj} = \operatorname{tr}(BA)$.

(iii) $\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(APP^{-1}) = \operatorname{tr}(A)$ by the cyclic property.

(iv) Over $\mathbb{C}$, A is similar to an upper triangular matrix T (Schur form), so $\operatorname{tr}(A) = \operatorname{tr}(T) = \sum \lambda_i$. Over $\mathbb{R}$ with A symmetric, use the spectral theorem.

(v) $\operatorname{tr}(A^T) = \sum_i (A^T)_{ii} = \sum_i A_{ii} = \operatorname{tr}(A)$.

(vi) $\operatorname{tr}(A^T B) = \sum_i (A^T B)_{ii} = \sum_i \sum_j A_{ji}B_{ji} = \sum_{i,j} A_{ij}B_{ij}$. □

Corollary 10.2. *For* $A \in \mathbb{F}^{n\times n}$ *with eigenvalues* $\lambda_1, \ldots, \lambda_n$*:* $\operatorname{tr}(A) = \sum_i \lambda_i$ *and* $\det(A) = \prod_i \lambda_i$.

Proof. The trace identity is Theorem 10.1(iv). For the determinant: $\det(A) = \det(P^{-1}TP) = \det(T) = \prod T_{ii} = \prod \lambda_i$ (where T is upper triangular with eigenvalues on the diagonal). □

10.2 Matrix Norms

Definition 10.2 (Operator Norm)**.** For $A \in \mathbb{R}^{m\times n}$ and a vector norm $\|\cdot\|$ on both $\mathbb{R}^n$ and $\mathbb{R}^m$, the *operator norm* (induced norm) is

$$\|A\|_{\text{op}} = \sup_{x\neq 0} \frac{\|Ax\|}{\|x\|} = \sup_{\|x\|=1} \|Ax\|.$$

For the Euclidean norm, $\|A\|_{\text{op}} = \sigma_1(A)$ (the largest singular value).

Definition 10.3 (Frobenius Norm)**.** $\|A\|_F = \sqrt{\sum_{i,j} A_{ij}^2} = \sqrt{\operatorname{tr}(A^T A)} = \sqrt{\sum_i \sigma_i^2}$.

Proposition 10.3 (Norm Relationships)**.** *For* $A \in \mathbb{R}^{m\times n}$ *with* $r = \operatorname{rank} A$*:*

$$\|A\|_{op} \leq \|A\|_F \leq \sqrt{r}\,\|A\|_{op}.$$

Proof. $\|A\|_{\text{op}} = \sigma_1 \leq \sqrt{\sigma_1^2 + \cdots + \sigma_r^2} = \|A\|_F$. For the upper bound: $\|A\|_F = \sqrt{\sum_{i=1}^{r} \sigma_i^2} \leq \sqrt{r \cdot \sigma_1^2} = \sqrt{r}\,\sigma_1 = \sqrt{r}\,\|A\|_{\text{op}}$. □

Theorem 10.4 (Submultiplicativity)**.** *For the operator norm,* $\|AB\|_{op} \leq \|A\|_{op}\|B\|_{op}$.

Proof. For any x with $\|x\| = 1$, $\|ABx\| \le \|A\|_{\text{op}}\|Bx\| \le \|A\|_{\text{op}}\|B\|_{\text{op}}$. Taking the supremum gives the result. □

Definition 10.4 (Condition Number)**.** For an invertible matrix A, the *condition number* (in the operator norm) is

$$\kappa(A) = \|A\|_{\text{op}} \cdot \|A^{-1}\|_{\text{op}} = \frac{\sigma_1}{\sigma_n},$$

where $\sigma_1 \ge \cdots \ge \sigma_n > 0$ are the singular values. For a symmetric positive definite matrix, $\kappa(A) = \lambda_{\max}/\lambda_{\min}$.

Proposition 10.5 (Condition Number and Sensitivity)**.** *Consider the linear system $Ax = b$ with A invertible. If b is perturbed to $b + \delta b$ and the solution changes from x to $x + \delta x$, then*

$$\frac{\|\delta x\|}{\|x\|} \le \kappa(A)\frac{\|\delta b\|}{\|b\|}.$$

Proof. $A(x + \delta x) = b + \delta b$ gives $\delta x = A^{-1}\delta b$, so $\|\delta x\| \le \|A^{-1}\|_{\text{op}}\|\delta b\|$. Also $\|b\| = \|Ax\| \le \|A\|_{\text{op}}\|x\|$, giving $1/\|x\| \le \|A\|_{\text{op}}/\|b\|$. Combining: $\|\delta x\|/\|x\| \le \|A^{-1}\|_{\text{op}}\|A\|_{\text{op}}\|\delta b\|/\|b\| = \kappa(A)\|\delta b\|/\|b\|$. □

10.3 The Matrix Exponential

Definition 10.5 (Matrix Exponential)**.** For $A \in \mathbb{R}^{n\times n}$, the *matrix exponential* is

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!},$$

which converges absolutely in any matrix norm (since $\sum \|A\|^k/k! = e^{\|A\|} < \infty$).

Proposition 10.6 (Properties of the Matrix Exponential)**.**

1. $e^0 = I$.
2. *If $AB = BA$, then $e^{A+B} = e^A e^B$.*
3. *e^A is always invertible, with $(e^A)^{-1} = e^{-A}$.*
4. $\frac{d}{dt}e^{tA} = Ae^{tA}$.

Proof. **(1)** $e^0 = \sum 0^k/k! = I$ (the $k = 0$ term).

(2) When $AB = BA$, the binomial theorem gives $(A + B)^k = \sum_{j=0}^{k} \binom{k}{j} A^j B^{k-j}$. Then $e^{A+B} = \sum_k \frac{1}{k!}\sum_j \binom{k}{j} A^j B^{k-j} = \sum_j \sum_\ell \frac{A^j}{j!}\frac{B^\ell}{\ell!} = e^A e^B$ (reindexing $\ell = k - j$ and using absolute convergence to rearrange).

(3) Since A and $-A$ commute, $e^A e^{-A} = e^{A-A} = e^0 = I$.

(4) $\frac{d}{dt}e^{tA} = \frac{d}{dt}\sum_k \frac{(tA)^k}{k!} = \sum_{k=1}^{\infty} \frac{kt^{k-1}A^k}{k!} = A\sum_{k=1}^{\infty} \frac{(tA)^{k-1}}{(k-1)!} = Ae^{tA}$. □

Proposition 10.7 (Matrix Exponential of Diagonalizable Matrices)**.** *If $A = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, then $e^A = P\operatorname{diag}(e^{\lambda_1}, \ldots, e^{\lambda_n})P^{-1}$.*

Proof. $A^k = PD^kP^{-1}$, so $e^A = \sum_k PD^kP^{-1}/k! = P(\sum_k D^k/k!)P^{-1} = Pe^DP^{-1}$. Since D is diagonal, $e^D = \operatorname{diag}(e^{\lambda_1}, \ldots, e^{\lambda_n})$. □

Matrix Exponential in Dynamical Systems

The matrix exponential solves the linear ODE $\dot{x}(t) = Ax(t)$: the solution is $x(t) = e^{tA}x(0)$. This appears in continuous-time Markov chains (where the transition semigroup is $P(t) = e^{tQ}$ for a rate matrix Q) and in the analysis of linear dynamical systems that arise in control theory (Phase 06). The condition number $\kappa(A)$ measures how sensitive these solutions are to perturbations in A.

11 Matrix Calculus

This section collects the derivative formulas needed for optimization and machine learning. All derivatives are taken in the *numerator layout* convention: if $f : \mathbb{R}^n \to \mathbb{R}$, then $\nabla f \in \mathbb{R}^n$ is a column vector.

Definition 11.1 (Gradient)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable. The *gradient* of f at x is the vector $\nabla f(x) \in \mathbb{R}^n$ with entries $(\nabla f)_i = \partial f/\partial x_i$. The gradient satisfies $f(x+h) = f(x) + \nabla f(x)^T h + o(\|h\|)$.

Definition 11.2 (Jacobian)**.** Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be differentiable. The *Jacobian* of f at x is the $m \times n$ matrix $J_f(x)$ with entries $(J_f)_{ij} = \partial f_i/\partial x_j$.

Definition 11.3 (Hessian)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice differentiable. The *Hessian* of f at x is the $n \times n$ matrix $\nabla^2 f(x)$ with entries $(\nabla^2 f)_{ij} = \partial^2 f/\partial x_i \partial x_j$. If f is C^2, the Hessian is symmetric (by equality of mixed partials).

Theorem 11.1 (Fundamental Derivative Formulas)**.** *Let $a, x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$.*

(i) $\nabla_x(a^T x) = a$.
(ii) $\nabla_x(x^T A x) = (A + A^T)x$. *If A is symmetric, this equals $2Ax$.*
(iii) $\nabla_x \|x\|^2 = 2x$.
(iv) $\nabla_x(x^T x) = 2x$ *(special case of (ii) with $A = I$).*

Proof. **(i)** $a^T(x+h) - a^T x = a^T h$, so the derivative is a.

(ii) $(x+h)^T A(x+h) - x^T A x = h^T A x + x^T A h + h^T A h$. The linear part is $h^T A x + x^T A h = h^T A x + (A^T x)^T h = (Ax + A^T x)^T h = ((A + A^T)x)^T h$. So $\nabla_x(x^T A x) = (A + A^T)x$.

(iii) and **(iv)** follow from (ii) with $A = I$. □

Theorem 11.2 (Derivatives Involving the Trace)**.** *Let $A \in \mathbb{R}^{m\times n}$ be a constant matrix and $X \in \mathbb{R}^{n\times m}$ a matrix variable.*

(i) $\frac{\partial}{\partial X} \operatorname{tr}(AX) = A^T$.
(ii) $\frac{\partial}{\partial X} \operatorname{tr}(X^T A) = A$.
(iii) $\frac{\partial}{\partial X} \operatorname{tr}(X^T A X) = (A + A^T)X$.

Here $\frac{\partial f}{\partial X}$ *is the matrix with entries* $\left(\frac{\partial f}{\partial X}\right)_{ij} = \frac{\partial f}{\partial X_{ij}}$.

Proof. **(i)** $\operatorname{tr}(AX) = \sum_{i,j} A_{ij} X_{ji}$, so $\frac{\partial}{\partial X_{k\ell}} \operatorname{tr}(AX) = A_{\ell k} = (A^T)_{k\ell}$.

(ii) $\operatorname{tr}(X^T A) = \operatorname{tr}(A^T X)$ (by $\operatorname{tr}(M) = \operatorname{tr}(M^T)$). Applying (i) with A replaced by A^T: $\frac{\partial}{\partial X} \operatorname{tr}(A^T X) = (A^T)^T = A$.

(iii) $\operatorname{tr}(X^T A X) = \sum_{i,j,k} X_{ji} A_{jk} X_{ki}$. Taking $\frac{\partial}{\partial X_{pq}}$: the terms where X_{pq} appears are those with $(j,i) = (p,q)$ (from the first X) and $(k,i) = (p,q)$ (from the second X). This gives $\sum_k A_{pk} X_{kq} + \sum_j X_{jq} A_{jp} = (AX)_{pq} + (A^T X)_{pq} = ((A + A^T)X)_{pq}$. □

Proposition 11.3 (Chain Rule for Compositions)**.** *If $g : \mathbb{R}^n \to \mathbb{R}^m$ and $f : \mathbb{R}^m \to \mathbb{R}$ are differentiable, then $\nabla_x(f \circ g)(x) = J_g(x)^T \nabla_y f(g(x))$.*

Proof. By the chain rule for multivariable functions, $\frac{\partial(f\circ g)}{\partial x_i} = \sum_{j=1}^m \frac{\partial f}{\partial y_j}\frac{\partial g_j}{\partial x_i}$. In matrix notation, this is $\nabla_x(f \circ g) = J_g^T \nabla_y f$. □

Matrix Calculus: The Toolbox for Gradient-Based Optimization

The formulas in this section are the computational backbone of gradient-based optimization in machine learning. The gradient of a loss function $\ell(\theta)$ with respect to parameters θ is computed by composing these elementary derivatives via the chain rule. In neural

networks, this composition is called *backpropagation*. In policy gradient methods, the gradient $\nabla_\theta J(\theta)$ involves derivatives of log-probabilities with respect to policy parameters.